COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures & Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	7/29/26	**DATE SUBMITTED:**	
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ACTIVITY NO 4.1 **Seatwork 4.1: Array-Based vs. STL Stack Implementations**

1. PROCEDURE OUTPUT

ILO A: Implement a stack using the C++ STL (std::stack).

```
std::stack<int> stack1, stack2;
std::cout << "Testing the stack STL:" << std::endl;

//isEmpty
std::cout << "is the stack empty? " << stack1.empty() << std::endl;
```

```
  //push
  stack1.push(10);
```

```
//top
std::cout << "The top of the stack is " << stack1.top() <<std::endl;

stack1.push(9);
std::cout << "The top of the stack is " << stack1.top() <<std::endl;

stack1.push(8);
std::cout << "The top of the stack is " << stack1.top() <<std::endl;

stack1.push(7);
std::cout << "The top of the stack is " << stack1.top() <<std::endl;
```

```
// pop
std::cout << "Testing pop:" << std::endl;
stack1.pop();
std::cout << "The top of the stack is " << stack1.top() <<std::endl;
```

```
// stack 1 size
std::cout << "The size of the stack is " << stack1.size() << std::endl;
```

Stack 2

```
//stack 2 creation
std::cout << "=====STACK 2====" << std::endl;
stack2.push(1);
std::cout << "The top of the stack is " << stack2.top() <<std::endl;

stack2.push(2);
std::cout << "The top of the stack is " << stack2.top() <<std::endl;

stack2.push(3);
std::cout << "The top of the stack is " << stack2.top() <<std::endl;

stack2.push(4);
std::cout << "The top of the stack is " << stack2.top() <<std::endl;

// stack 2 size
std::cout << "The size of the stack is " << stack2.size() << std::endl;
```

```cpp
// swap
std::cout << "Testing swap:" << std::endl;
stack1.swap(stack2);


//display stack1
std::cout << "stack1: ";
    while (!stack1.empty()) {
        std::cout << stack1.top() << " ";
        stack1.pop();
}

std::cout << std::endl;

//display stack2
std::cout << "stack2: ";
    while (!stack2.empty()) {
        std::cout << stack2.top() << " ";
        stack2.pop();
}



return 0;
```

```
Testing the stack STL:
is the stack empty? 1
=====STACK 1====
The top of the stack is 10
The top of the stack is 9
The top of the stack is 8
The top of the stack is 7
Testing pop:
The top of the stack is 8
is the stack empty? 0
The size of the stack is 3
Testing Emplace:
The top of the stack is 67
=====STACK 2====
The top of the stack is 1
The top of the stack is 2
The top of the stack is 3
The top of the stack is 4
The size of the stack is 4
Testing swap:
stack1: 4 3 2 1
stack2: 67 8 9 10
```

The stack program showed how different STL stack functions work. At first, the stack was empty, then values were added using push() and emplace(), removed using pop(), and the top element was viewed using top(). The activity also showed that stacks follow the Last In, First Out (LIFO) principle, where the last value added is the first value

removed. Finally, the swap() function successfully exchanged the contents of the two stacks since I created stack2 so I can use the function.

ILO B.1: Implement a stack using a Static Array.

```
#include <iostream>

//global declarations
#define maxCap 10

int stackArr[maxCap];
int topIndex = -1;
int newData;

//prototype functions
void push();
void pop();
void Top();
bool isEmpty();
bool isFull();
void displayStack();
```

```
int main() {
// maindriver
    int choice;

    while(true)
    {
        std::cout << "-----------------------------------------" <<std::endl;
        std::cout << "Stack Operation:" << std::endl;
        std::cout <<"1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display" << std::endl;
        std::cout << "-----------------------------------------" <<std::endl;
        std::cin >> choice;
        std::cout << std::endl;

        switch(choice){
            case 1: push();
            break;

            case 2: pop();
            break;

            case 3: Top();
            break;

            case 4: std::cout << "Is stack empty? " << isEmpty() << std::endl;
            break;

            case 5: std::cout << "Is stack full? " << isFull() << std::endl;
            break;

            case 6: displayStack();
            break;

            default: std::cout << "Invalid Choice" << std::endl;
            break;
        }

    }
```

```
//function definition
bool isEmpty() {
    if(topIndex == -1) return true;
    return false;
}

bool isFull() {
    if(topIndex == maxCap -1) return true;
    return false;
}

    //error checking
void push() {
    if(isFull()){
        std::cout << "Stack overflow" << std::endl;
        return;
    }

    //pushing to the stack
    std::cout << "Enter a new value: ";
    std::cin >> newData;
    stackArr[++topIndex] = newData;
}

void pop() {
    if(isEmpty()) {
        std::cout << "Stack underflow" << std::endl;
        return;
    }
    //display the value we are going to pop
    std::cout << "Popping: " << stackArr[topIndex] << std::endl;
    //decrement the top value from the stack
    topIndex--;

}
```

```
void displayStack() {
    if(isEmpty()) {
        std::cout << "Stack is empty" << std::endl;
        return;
    }
    std::cout << "Stack Elements:" << std::endl;
    for (int i = topIndex; i >= 0; i--) {
        std::cout << stackArr[i] << " ";
    }
    std::cout << std::endl;

}
```

```
Stack Operation:
1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display
==========================================
1

Enter a new value: 1
==========================================
Stack Operation:
1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display
==========================================
1

Enter a new value: 2
==========================================
Stack Operation:
1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display
==========================================
1

Enter a new value: 3
==========================================
Stack Operation:
1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display
==========================================
6

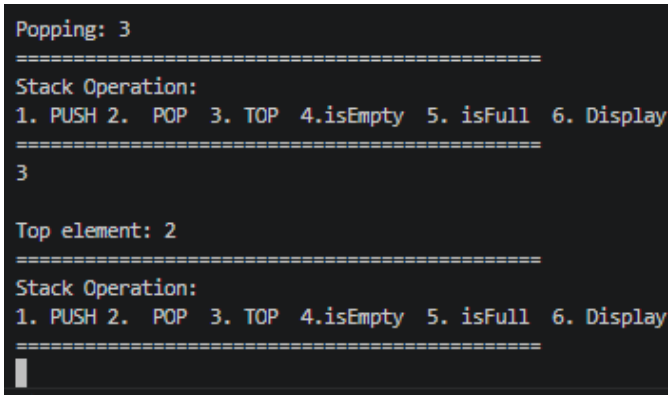
Stack Elements:
3 2 1
```

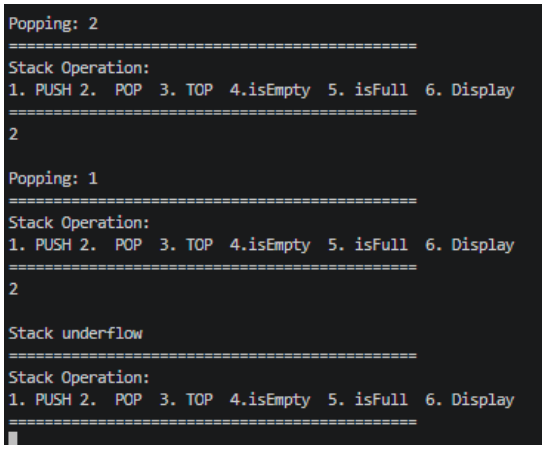
```
Stack Operation:
1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display
==========================================
4

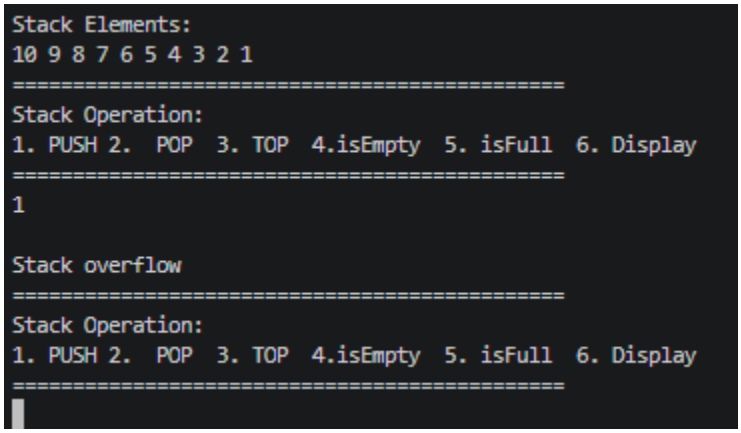
Is stack empty? 0
==========================================
Stack Operation:
1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display
==========================================
5

Is stack full? 0
==========================================
Stack Operation:
1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display
==========================================
3

Top element: 3
==========================================
Stack Operation:
1. PUSH 2.  POP  3. TOP  4.isEmpty  5. isFull  6. Display
==========================================
```







The stack program showed how the different stack operations work using an array implementation. At first, the stack was empty, then values were added using push(), removed using pop(), and viewed using Top(). It also has overflow and underflow checking and it displays the elements with the LIFO format.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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